

# Story of Arithmetic

Part 2





## About the Course

This course gives students a glimpse into how the ideas that guide our study of arithmetic were first discovered.



## Placement & Combining Tips

This course is intended to follow Part 1; however, the order may be reversed to facilitate combining students.



## Scheduling

| GRADE | SCHEDULE INFO.                                    | BOOKS                                                                                              |
|-------|---------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 6     | Story of Arithmetic: Part 2<br>1 time/week 15 min | Number Stories of Long Ago<br>Historical Atlas of the World 6th Ed.<br>Student World Atlas 6th Ed. |

### Sample Weekly View

| Day 1                       | Day 2                       | Day 3 | Day 4 | Day 5 |
|-----------------------------|-----------------------------|-------|-------|-------|
| Story of Arithmetic: Part 2 |                             |       |       |       |
|                             | Story of Arithmetic: Part 2 |       |       |       |



## Planning & Prep

Permission to print for non-commercial use. See Alveary group use policy to use lessons in a group context.

**LINKS:** Click text or scan the QR code in the top corner of the lesson plan pages to view online resources associated with the lessons.

Responsibility for previewing all links rests with the teacher. All links were checked at the time of publication; however, websites change frequently and may contain objectionable content. Please report broken links by contacting us through our website.

Story of Arithmetic: Part 2

☐ Term 1: Familiarize yourself with the "lattice method." (See Lesson 6.)



## Books & Resources

For book rationales and purchase options, click the Book List link or scan the QR code below.

∞ [View Book List Details](#)

## Story of Arithmetic: Part 2



Number Stories of Long Ago



Historical Atlas of the World 6th Ed.



Student World Atlas 6th Ed.



## Supplies

For supply list details and basic supplies helpful to have on hand, click the links or scan the QR code below.

∞ [View Basic Supplies](#)

(No Subject Supplies Assigned)



## Quick Links

(No Quick Links Assigned)

Click THIS text  
or scan the QR  
code for links.



# Story of Arithmetic: Part 2

## How To Teach



### Recap

Connect back to previous lesson.



### Introduce

Optional: Use a picture, map, or engaging question to prepare the student to receive the idea from the day's lesson. Talk about one or two unknown words that will be crucial to understanding.



### Read

Read the new passage out loud or have the student read it orally or silently.



### Narrate

Have the student narrate orally or in writing.



### Discuss

- Give students an opportunity to share their thoughts on the lesson and any connections made.



### Practice

Optional: Attempt to recreate some of the problems from the stories.

# Story of Arithmetic: Part 2

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## Term 1

### WEEK 1 15m Story of Arithmetic: Part 2 - Lesson 1

*Multiplication: Introduction*

Materials: Number Stories of Long Ago

#### → INTRO

We left off when people had learned to add and subtract pretty easily, but multiplication and division were a problem. This year, we will see how these methods were developed. We begin in the 1100s. "Moor" is an old word for a Muslim person from regions in southern Spain and northern Africa. We don't use it anymore.

#### → READ, NARRATE, & DISCUSS

p.65-66 "So you wonder" - "Leonardo and Johann."

### WEEK 2 15m Story of Arithmetic: Part 2 - Lesson 2

*Multiplication: Leonardo of Pisa*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.66-67 "When Leonardo of" - "we are taught."

#### → PRACTICE

Try multiplying  $36 \times 5$  Leonardo's way.

### WEEK 3 15m Story of Arithmetic: Part 2 - Lesson 3

*Multiplication: Tonstall and Widman*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → INTRO

Today, you will learn about two more boys and their style of multiplication.

#### → READ, NARRATE, & DISCUSS

p.67-71 "It was about" - "Leonardo and Johann."

### WEEK 4 15m Story of Arithmetic: Part 2 - Lesson 4

*Multiplication: The Grating Plan*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

# Story of Arithmetic: Part 2

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| Term 1                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                  |
|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>p.71-72 "I don't see" - "answered the Story-Teller."</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                                                                                                  |
| <b>WEEK 5</b>                                                                                            | <p> <b>15m Story of Arithmetic: Part 2 - Lesson 5</b></p> <p><i>Practice: Cuthbert's Method</i></p> <div> Materials: Number Stories of Long Ago</div> <p>→ <b>RECAP</b></p> <p>What did you read last time?</p> <p>→ <b>STUDY</b></p> <p>Study the pictures. The problem is <math>934 \times 314</math>. Work it out our way, and then look again at the pictures. Can you figure out how this method works?</p> <p> Image Resource: Number Stories of Long Ago "From the First Printed Arithmetic" p.70</p>                                                                                                |                                                                                                                  |
| <b>WEEK 6</b>                                                                                            | <p> <b>15m Story of Arithmetic: Part 2 - Lesson 6</b></p> <p><i>Practice: The Lattice Method</i></p> <div> Materials: Number Stories of Long Ago</div> <p>→ <b>VIEW</b></p> <p>∞ Video Link: Lattice Multiplication (6:12)</p> <p>→ <b>PRACTICE</b></p> <p>Look again at the pictures. Use the lattice method to multiply <math>934 \times 314</math>. Which of the four methods shown matches your work? Make up another problem and use the lattice method to solve it. Check your work using our method.</p> <p> Image Resource: Number Stories of Long Ago "From the First Printed Arithmetic" p.70</p> |                                                                                                                  |
| <b>WEEK 7</b>                                                                                            | <p> <b>15m Story of Arithmetic: Part 2 - Lesson 7</b></p> <p><i>Division: Scratch and Galley Methods</i></p> <div> Materials: Number Stories of Long Ago</div> <p>→ <b>READ, NARRATE, &amp; DISCUSS</b></p> <p>p.75-77 "Since this is" - "learned in school."</p> <p>→ <b>STUDY</b></p> <p>Look at the image. Can you find 1728 and 12? How many times will 12 go into 17? Work the problem our way. Can you find where Filippo recorded his answer?</p> <p> Image Resource: Number Stories of Long Ago "How Filippo Proved His Work" p.76</p>                                                              | <p>• <b>IMPORTANT DATES</b></p> <p>1st Arithmetic Bk, England (1522)</p> <p>1st Arithmetic Bk, Mexico (1556)</p> |

# Story of Arithmetic: Part 2

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## Term 1

### WEEK 8 15m Story of Arithmetic: Part 2 - Lesson 8

*Division: Filippo's New Method*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.77-80 "After Filippo had" - "still more simple."

### WEEK 9 15m Story of Arithmetic: Part 2 - Lesson 9

*Division: Adriaen and Michael*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.80 "When Filippo became" - "two centuries later."

#### → STUDY

"How Adriaen Divided." Can you find the 150 and the 6? How many times will 6 go into 15? [2 with 3 left over] Where did Adriaen record the 2 and the 3? Can you figure out the rest?

### WEEK 10 15m Story of Arithmetic: Part 2 - Lesson 10

*Division: Conclusion*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.81-82 "If you had" - "laughed the Crowd."

### WEEK 11 15m Story of Arithmetic: Part 2 - Lesson 11

*Practice*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → PRACTICE

How many 5's will go into 565? Solve using manipulatives first and then with long division.

#### → VIEW

# Story of Arithmetic: Part 2

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| Term 1                                                                                                                                                                                                                                                                                                                   |  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| <p>This teacher is showing us three different algorithms for solving the same problem. An algorithm is a sort of shortcut for getting an answer without using real objects. Which do you think is easiest? Why?</p> <p>∞ Video Link: Three Methods of Division (5:10)</p>                                                |  |
| <p><b>WEEK 12</b>  <b>15m Story of Arithmetic: Part 2 - Lesson 12</b></p> <p><i>Exam Week</i></p> <p><b>→ EXAMS</b></p> <p>Answer question(s) related to course.</p> <p>Questions will come from:</p> <p>Number Stories of Long Ago</p> |  |



# Story of Arithmetic: Part 2

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## Term 2

### WEEK 1 15m Story of Arithmetic: Part 2 - Lesson 13

*Fractions: Introduction*

Materials: Number Stories of Long Ago

#### → RECAP

What do you remember from last term?

#### → READ, NARRATE, & DISCUSS

p.85-86 "Now we are" - "the Story-Teller began:"

### WEEK 2 15m Story of Arithmetic: Part 2 - Lesson 14

*Fractions: Ahmes of Egypt*

Materials: Number Stories of Long Ago, Historical Atlas of the World

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.86-87 "When Ahmes had" - "the Christian era."

#### → LOCATE

Map Resource: Historical Atlas of the World "International Trade 1350-1450 A.D." p.28-29

Find Alexandria. What can you tell about this city by looking at the map?

### WEEK 3 15m Story of Arithmetic: Part 2 - Lesson 15

*Fractions: Heron and Jakob*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → INTRO

Look at an analog clock. Can you discover how it shows fractions? What about a digital clock?

#### → READ, NARRATE, & DISCUSS

p.87-88 "When Heron went" - "else thought so."

#### → DISCUSS

Have you ever heard someone complain about "new-fangled math" like Jakob's teacher? How might you answer that person, now that you have studied some of the history of math?

# Story of Arithmetic: Part 2

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## Term 2

### WEEK 4 15m Story of Arithmetic: Part 2 - Lesson 16

*Fractions: Simon of Belgium*

Materials: Number Stories of Long Ago, Student World Atlas

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.89-91 "It was nearly" - "started for bed."

#### → LOCATE

Map Resource: Student World Atlas "Political Map of Europe" p.83  
Belgium

#### → DISCUSS

What did you think when you read about "fashions" in arithmetic? Do you think other subjects have fashions, too?

### WEEK 5 15m Story of Arithmetic: Part 2 - Lesson 17

*Number Puzzles: Introduction*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.93-95 "So this is" - "of the ages."

### WEEK 6 15m Story of Arithmetic: Part 2 - Lesson 18

*Number Puzzles: Ching and Menes*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.95-98 "In my land" - "secrets of arithmetic."

#### → PRACTICE

Test Menes's theory that if you take the difference of a number and its reverse, this difference is always divisible by 9.

# Story of Arithmetic: Part 2

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## Term 2

### WEEK 7 15m Story of Arithmetic: Part 2 - Lesson 19

*Number Puzzles: Heron and Gerbert*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.98-99 "One of the" - "do the trick?"

#### → STUDY

The box of numbers. Can you figure out how Adriaen chose his numbers so that the sum would be 29,997? [Try adding  $2376 + 7623$  and  $4152 + 5847$ .

Can you figure out the rest?]

Image Resource: Number Stories of Long Ago p.98-99

#### • IMPORTANT DATES

Heron of Alexandria (10 AD-70 AD)

### WEEK 8 15m Story of Arithmetic: Part 2 - Lesson 20

*Word Puzzles*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.99-101 "I think," said" - "it for you."

#### → PRACTICE

Test at least one of the tricks in this section.

### WEEK 9 15m Story of Arithmetic: Part 2 - Lesson 21

*More Number Puzzles*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.101-103 "Can any of" - "around the room."

#### → PRACTICE

Test at least one of the tricks in this section.

# Story of Arithmetic: Part 2

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## Term 2

**WEEK 10** 15m Story of Arithmetic: Part 2 - Lesson 22

*Puzzles to Solve*

Materials: Number Stories of Long Ago

→ **RECAP**

What did you read last time?

→ **READ, NARRATE, & DISCUSS**

p.103-105 "Yes," said Menes" - "you do so?"

→ **PRACTICE**

Test at least one of the tricks in this section.

**WEEK 11** 15m Story of Arithmetic: Part 2 - Lesson 23

*Catch-Up Day*

→ **CATCH UP**

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

**WEEK 12** 15m Story of Arithmetic: Part 2 - Lesson 24

*Exam Week*

→ **EXAMS**

Answer question(s) related to course.

Questions will come from:  
Number Stories of Long Ago

# Story of Arithmetic: Part 2

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## Term 3

### WEEK 1 15m Story of Arithmetic: Part 2 - Lesson 25

*Curious and Strange Math Problems*

Materials: Number Stories of Long Ago

#### → RECAP

What do you remember from last term?

#### → READ, NARRATE, & DISCUSS

p.105-107 "I ran across" - "What is it?"

#### → PRACTICE

Test at least one of the tricks in this section.

### WEEK 2 15m Story of Arithmetic: Part 2 - Lesson 26

*Problems with Letters, Weight & Money*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.107-108 "Here is a" - "so he stopped."

#### → PRACTICE

Test at least one of the tricks in this section.

### WEEK 3 15m Story of Arithmetic: Part 2 - Lesson 27

*Conclusion of Number Puzzles*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.109-110 "This problem," said - "the log cabin."

#### → PRACTICE

Test at least one of the tricks in this section.

### WEEK 4 15m Story of Arithmetic: Part 2 - Lesson 28

*Two Story Problems*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

# Story of Arithmetic: Part 2

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## Term 3

p.113-115 "Have you imagination" - "of the boys."

### → PRACTICE

Solve the problems in this section.

## WEEK 5 15m Story of Arithmetic: Part 2 - Lesson 29

*More Story Problems*

Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.115-116 "I will give" - "that for ourselves."

### → PRACTICE

How much did it cost to shoe the horse?

## WEEK 6 15m Story of Arithmetic: Part 2 - Lesson 30

*Even More Story Problems*

Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.116-118 "Here is an" - "to find out."

### → PRACTICE

Solve one of the problems in this section.

## WEEK 7 15m Story of Arithmetic: Part 2 - Lesson 31

*Just a Few More Story Problems*

Materials: Number Stories of Long Ago

### → RECAP

What did you read last time?

### → READ, NARRATE, & DISCUSS

p.118-119 "It is getting" - "ahead of Achilles."

### → PRACTICE

Solve one of the problems in this section.

# Story of Arithmetic: Part 2

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## Term 3

### WEEK 8 15m Story of Arithmetic: Part 2 - Lesson 32

*More Math Problems*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.119-121 "I don't know" - "to do it?"

#### → PRACTICE

Solve one of the problems in this section.

### WEEK 9 15m Story of Arithmetic: Part 2 - Lesson 33

*Conclusion of Puzzles*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → READ, NARRATE, & DISCUSS

p.121-123 "Here are two" - "with the Memories."

#### → PRACTICE

Solve one of the problems in this section.

### WEEK 10 15m Story of Arithmetic: Part 2 - Lesson 34

*Practice*

Materials: Number Stories of Long Ago

#### → RECAP

What did you read last time?

#### → PRACTICE

Answer question #4 "The Question Box" p.124

### WEEK 11 15m Story of Arithmetic: Part 2 - Lesson 35

*Catch-Up Day*

#### → CATCH UP

Use this time to catch up on lessons in this or another course, or explore Extra Helpings.

### WEEK 12 15m Story of Arithmetic: Part 2 - Lesson 36

*Exam Week*

#### → EXAMS

Answer question(s) related to course.

# Story of Arithmetic: Part 2

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## Term 3

Questions will come from:  
Number Stories of Long Ago



# Story of Arithmetic: Part 2

## Examination

- Term 1** · Number Stories of Long Ago: Tell what you know about how methods of multiplication have changed over time.
- Term 2** · Number Stories of Long Ago: Tell what you know about how fractions have changed over time. OR Tell about one of the number puzzles you found interesting.
- Term 3** · Number Stories of Long Ago: Tell about a story problem or puzzle you found interesting this term.